



Deneshwaran M

AI/ML ENGINEER | DATA SCIENCE

✉ reach.deneshwaran@gmail.com

📞 +91 7826936660

📍 Madurai, Tamil Nadu

🌐 <https://www.linkedin.com/in/deneshwaran21/>



PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in building and deploying production-grade machine learning models. Experienced in supervised/unsupervised learning, deep learning, and Generative AI including LLM integrations and RAG architectures. Proficient in Python, SQL, and statistical modeling, with hands-on MLOps experience using Azure ML, Docker, and MLflow. Strong full-stack foundation (APIs, databases, cloud) enables seamless end-to-delivery of AI-powered products from data ingestion to model inference. Passionate about translating complex business problems into data-driven, measurable AI solutions that drive ROI. Currently expanding expertise in Azure AI Stack & MLOps.



PROFESSIONAL EXPERIENCE

Junior Software Engineer

Vhilv Technology Private Limited Sep 2025 - Jun 2026

- Spearheaded development of 10+ intelligent UI components and high-performance backend microservices for NAS OS, leveraging React, Node.js, and RESTful APIs. Engineered advanced caching layers, optimized data retrieval pipelines, and implemented lazy-loading strategies reducing page load latency by 30% and improving model inference response times.
- Orchestrated end-to-end feature delivery as core member of 5-member Agile squad actively participating in sprint planning, daily stand-ups, and retrospectives. Successfully rolled out 3 high-impact features ahead of schedule while maintaining 100% code coverage and zero critical bugs in production. Integrated real-time telemetry & performance monitoring to track feature adoption and system health.
- Streamlined internal development workflows by establishing code review checklists, task prioritization frameworks, and automated testing protocols. Collaborated with team members to guarantee 100% on-time delivery of all project milestones and sprint goals. Introduced data-driven decision-making for feature prioritization using user analytics & A/B testing methodologies.

Software Developer

Priyonix Global Ventures Pvt Ltd Jul 2025 - Aug 2025

- Engineered highly responsive, mobile-first web applications utilizing Next.js, React.js, and Tailwind CSS, implementing server-side rendering (SSR) and static site generation (SSG) to deliver blazing-fast, SEO-optimized user interfaces.
- Architected a scalable, atomic-design component library featuring 15+ reusable UI elements with dynamic prop-driven variants. This modular approach accelerated new feature velocity by 40% and directly boosted Core Web Vitals by an average of 20%, significantly enhancing end-user experience.



KEY COMPETENCY AREAS

- Python Data Stack
- Supervised & Unsupervised Learning
- End-to-End ML System Architecture & Model Deployment
- Model Interpretability (SHAP, LIME) & Data-Driven Decision Making
- Interactive Dashboards (Plotly, Tableau, Power BI) & KPI Tracking
- MLOps Pipelines
- Model Optimization
- Git, GitHub & CI/CD
- Problem Solving & Analytical Thinking



SKILLS

- Programming & Data: Python, SQL, Pandas, NumPy, PySpark, Matplotlib, Seaborn, Plotly
- Big Data & Streaming: Apache Kafka, Snowflake, Azure Data Factory, Azure Data Lake
- Statistics & Machine Learning: Statistics, Probability, Exploratory Data Analysis (EDA), Feature Engineering, Scikit-learn, XGBoost, Model Evaluation, Cross-Validation, Hyperparameter Tuning, A/B Testing
- Deep Learning & AI: PyTorch, TensorFlow, Keras, CNNs, Transfer Learning, Computer Vision, NLP, OpenCV
- Generative AI: LLMs, Prompt Engineering, RAG Architectures, LangChain, LangGraph, LLM APIs (OpenAI, Azure OpenAI)
- Explainable AI: SHAP, LIME
- MLOps & Deployment: FastAPI, REST APIs, Streamlit, Azure ML, MLflow
- Databases & Analytics: MongoDB, SQLite, PostgreSQL, Snowflake, Power BI, Excel, Tableau

- Seamlessly integrated complex RESTful APIs into the frontend ecosystem, implementing robust asynchronous state management, graceful error handling, and skeleton loading screens to ensure a smooth, glitch-free experience across all device viewports (Desktop, Tablet, Mobile).



PROJECTS

Contactless Fingerprint Quality Assessment & Scoring Pipeline

AI/ML Developer Aug 2026

- Developed an end-to-end automated Image Quality Control (QC) pipeline for contactless mobile fingerprint authentication using OpenCV, NumPy, and Scikit-image. Engineered 5 multi-metric quality assessment algorithms (Blur via Laplacian Variance, Brightness via Grayscale Intensity, Glare via Pixel Thresholding, ROI via Otsu Thresholding, Ridge Clarity via Gabor Filter Response) with weighted composite scoring (0–100) and dynamic pass/reject decision logic achieving >95% rejection accuracy for poor-quality captures.
- Built an interactive Streamlit dashboard with real-time image upload, dynamic threshold sliders, normalized metric visualizations, and actionable user guidance (e.g., *"Too blurry—hold steady," "Glare detected—tilt finger"*). Also developed batch evaluation pipeline processing 20 test images (5 per condition: good, blurry, dark, glare) with automated CSV report generation for validation and performance analysis.
- Optimized pipeline performance achieving sub-300ms total inference time (Blur <10ms, Brightness <5ms, Glare <10ms, ROI <100ms, Ridge <150ms) through efficient algorithm design. Containerized application using Docker with modular architecture enabling easy integration into biometric systems reducing downstream processing costs by ~40% by eliminating wasted compute on low-quality images.
- **Tech Stack:** Python, OpenCV, NumPy, Scikit-image, Streamlit, FastAPI, Pandas, Git
- [Live Link Github Repo](#)

Real-Time Waste Classification System using Transfer Learning

AI/ML Developer Jul 2026

- Engineered a custom ResNet-50 image classifier using PyTorch and Transfer Learning, fine-tuning on a custom 5-category dataset to achieve 95.96% validation accuracy for real-time waste material identification.
- Built and deployed a production-grade web application on Streamlit Cloud, integrating a FastAPI inference pipeline with a modern dark UI, enabling instantaneous image preprocessing and high-confidence predictive feedback.
- Solved the critical "information to action" gap by dynamically linking AI classification results to a local recycler database, transforming the classifier into a complete sustainability solution; delivered the functional MVP independently within a 7-day competitive hackathon.
- **Tech Stack:** PyTorch, ResNet-50, FastAPI, Streamlit, OpenCV, NumPy, Pandas, Scikit-learn
- [Live Link Github Repo](#)

- Development: Git, GitHub, CI/CD, JavaScript, React.js, Node.js



KEY TECHNICAL TRAININGS

Full Stack Web Development
Nxtwave

Data Analytics Foundation
NxtWave



ACHIEVEMENTS

Coding Platforms:

100+ CodeChef | 50+ LeetCode | 50+ HackerRank problems solved



EDUCATION

Bachelor of Technology (B.Tech) - Information Technology

J.J. College of Engineering and Technology

Nov 2021 - Jun 2025

CGPA: 8.86 / 10